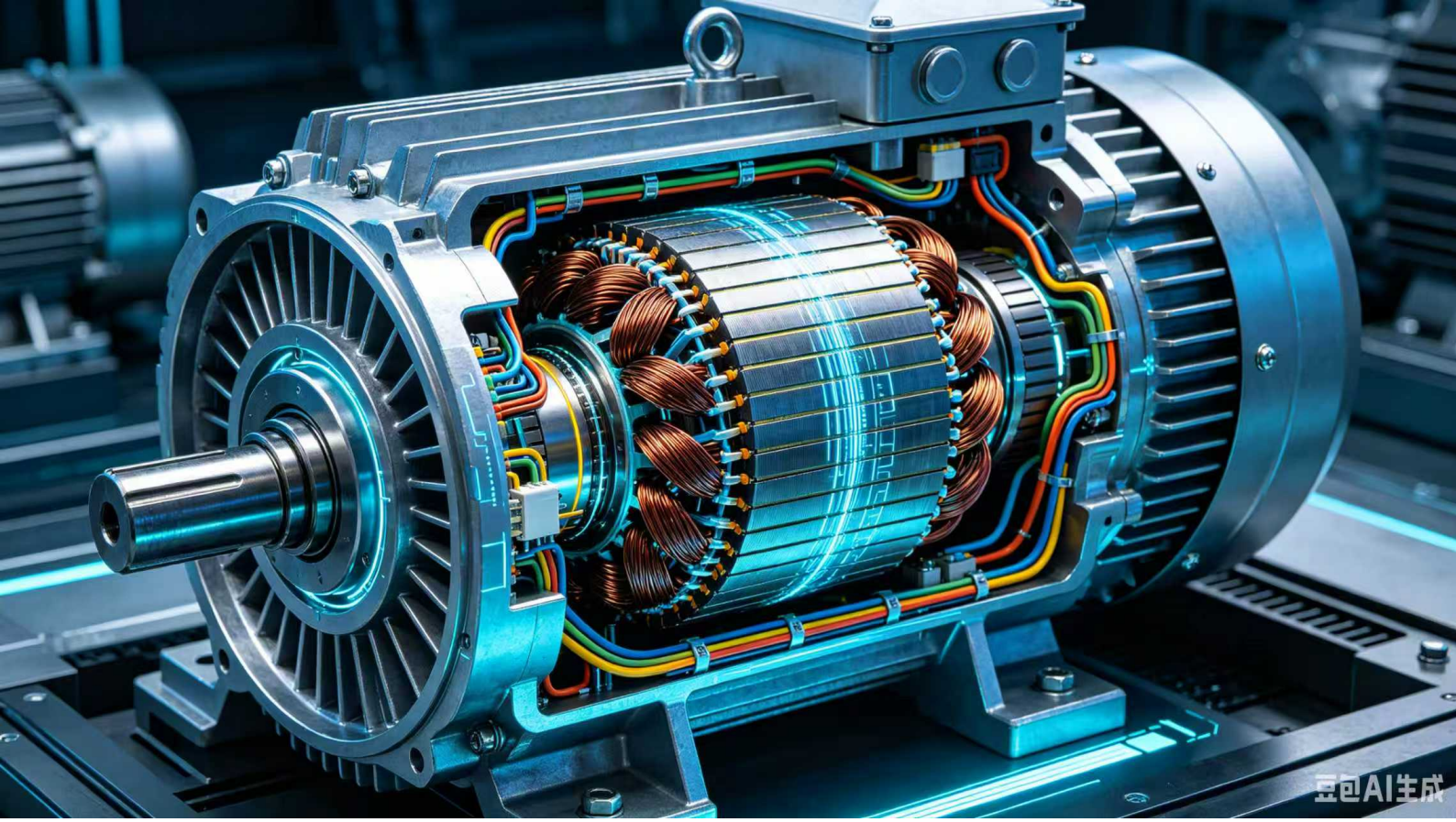


14. Maxwell's Equations and EM Waves

Nothing in life is to be feared, it is only to be understood.
– *attributed to Marie Curie*

Now is the time to understand more, so that we may fear less.
– *Glenn T. Seaborg (1967)*

14.1 Maxwell's Equations

Mathematically, we know that the divergence of a curl is always zero, namely, $\nabla \cdot (\nabla \times \mathbf{A}) = 0$. Let's examine this identity with $\mathbf{E}$ and $\mathbf{B}$.

For $\mathbf{E}$, we have:

$$\nabla \cdot (\nabla \times \mathbf{E}) = \nabla \cdot \left(-\frac{\partial \mathbf{B}}{\partial t} \right) = -\frac{\partial}{\partial t} (\nabla \cdot \mathbf{B})$$

which must be zero according to Gauss' law for magnetism.

However, for $\mathbf{B}$, we have:

$$\nabla \cdot (\nabla \times \mathbf{B}) = \nabla \cdot (\mu_0 \mathbf{j}) = \mu_0 \nabla \cdot \mathbf{j}$$

This is not necessarily zero if the current is not steady. Due to the continuity equation, $\nabla \cdot \mathbf{j} = -\frac{\partial \rho}{\partial t}$, therefore if the charge distribution varies with time, the above result violates the identity.

Maxwell pointed out that the identity can be restored, if an extra term is added to Ampère's law, namely:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j} + \mu_0 \mathbf{j}_d = \mu_0 \left(\mathbf{j} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)$$

In this way,

$$\begin{aligned} \nabla \cdot (\nabla \times \mathbf{B}) &= \mu_0 \nabla \cdot \left(\mathbf{j} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \\ &= \mu_0 \left(\nabla \cdot \mathbf{j} + \epsilon_0 \frac{\partial}{\partial t} \nabla \cdot \mathbf{E} \right) \\ &= \mu_0 \left(\nabla \cdot \mathbf{j} + \frac{\partial \rho}{\partial t} \right) \\ &= 0 \end{aligned}$$

The extra term $\mathbf{j}_d = \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$ is known as the **displacement current** density. With this addition, we have completed Maxwell's equations. Here S is a closed surface bounding the volume V in the first and third equations, and a surface bounded by the loop C in the second and fourth; in those last two, the loop and the surface it bounds are taken to be at rest. The equations now read:

$$\begin{aligned} \oint_S \mathbf{E} \cdot d\mathbf{S} &= \frac{1}{\epsilon_0} \int_V \rho \, dV \\ \oint_C \mathbf{E} \cdot d\mathbf{l} &= -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{S} \\ \oint_S \mathbf{B} \cdot d\mathbf{S} &= 0 \\ \oint_C \mathbf{B} \cdot d\mathbf{l} &= \mu_0 \int_S \mathbf{j} \cdot d\mathbf{S} + \mu_0 \epsilon_0 \frac{d}{dt} \int_S \mathbf{E} \cdot d\mathbf{S} \end{aligned} \tag{14.1}$$

or in the differential form, where ρ and $\mathbf{j}$ include all charges and currents:

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0} \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \cdot \mathbf{B} &= 0 \\ \nabla \times \mathbf{B} &= \mu_0 \left(\mathbf{j} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \end{aligned} \tag{14.2}$$

The physical meaning of this additional term can be illustrated with a charging capacitor.

Recall that we explicitly pointed out that there are an infinite number of surfaces bounded by a specific loop C when we discussed Ampère's law. Now suppose we are charging up a parallel plate capacitor with a pair of straight wires. There is a current i in the wire. If we put a closed loop C around one of the wires, we immediately know that if we choose the surface that cuts the wire, Ampère's integral yields $\mu_0 i$. However, we can deform the surface, so that it encloses one of the plates and does not cut the wire (Fig. 14.1). If we stick to the original form of Ampère's law, this becomes a paradox. However, the displacement current term fixes this problem. Between the plates, with no fringing, $E = q/\epsilon_0 A$, where A is the area of a plate and q the charge on the positive plate; the displacement current through S_2 is then $\epsilon_0 \, d(EA)/dt = dq/dt = i$, exactly the conduction current through S_1.

In addition, now if we choose the loop between the plates of the charging capacitor, we immediately see that there is a magnetic field even though there is no conduction current. Just as a

changing magnetic field induces an electric field according to Faraday's law, Maxwell's fix states that a changing electric field also induces a magnetic field.

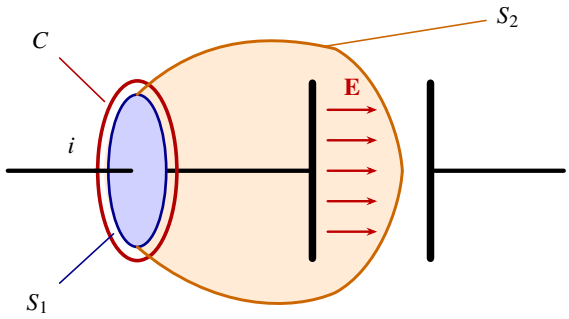


Figure 14.1: The displacement current illustrated with a charging parallel-plate capacitor. Two surfaces S_1 (cutting the wire) and S_2 (passing between the plates) are bounded by the same Amperian loop C. Without the displacement current term, Ampère's law gives contradictory results. The changing electric flux between the plates produces a displacement current $i_d = \epsilon_0 \, d\Phi_E/dt$ that exactly equals the conduction current i in the wire, resolving the paradox.

A logical perspective. Maxwell's equations can be organized by their physical roles:

$$\textbf{Constraints at each instant:}$$
$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \qquad \nabla \cdot \mathbf{B} = 0 \quad \text{(no magnetic charge)}$$
$$\textbf{How the fields change:}$$
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \qquad \nabla \times \mathbf{B} = \mu_0 \left(\mathbf{j} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)$$
$$\textbf{Fields affect charges:}$$
$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

Together with Newton's laws for the charges, these equations summarize the entire theoretical content of classical electrodynamics.

14.2 Magnetic Charge

Maxwell's equations display a striking asymmetry: there exist charges on which the field lines of $\mathbf{E}$ begin and end, but no magnetic charges for $\mathbf{B}$. In a sense, the equations "beg" for magnetic charge to exist. Paul Dirac showed that the existence of even a single magnetic monopole would explain why electric charge is quantized. However, despite extensive searches, no magnetic monopole has ever been conclusively observed. As far as we know, magnetic charge and magnetic current are zero everywhere. In Maxwell's theory $\nabla \cdot \mathbf{B} = 0$ holds exactly, and no experiment has confirmed a magnetic charge.

14.3 The Traveling Electromagnetic Wave

We now try to solve Maxwell's equations (14.2) in vacuum, that is, in a region with no charges or currents. There both ρ and $\mathbf{j}$ are zero, and Maxwell's equations become very symmetric:

$$\begin{aligned} \nabla \cdot \mathbf{E} &= 0 \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \cdot \mathbf{B} &= 0 \\ \nabla \times \mathbf{B} &= \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \end{aligned}$$

They can be decoupled by:

$$\begin{aligned} \nabla \times (\nabla \times \mathbf{E}) &= \nabla \times \left(-\frac{\partial \mathbf{B}}{\partial t} \right) \\ &= -\frac{\partial}{\partial t} (\nabla \times \mathbf{B}) \\ &= -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} \end{aligned}$$

and

$$\begin{aligned} \nabla \times (\nabla \times \mathbf{B}) &= \nabla \times \left(\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \\ &= \mu_0 \epsilon_0 \frac{\partial}{\partial t} (\nabla \times \mathbf{E}) \\ &= -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2} \end{aligned}$$

Noticing that

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$$

for any vector field $\mathbf{A}$, we have:

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\nabla^2 \mathbf{E}$$

and similarly $\nabla \times (\nabla \times \mathbf{B}) = -\nabla^2 \mathbf{B}$. We arrive at:

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

and

$$\nabla^2 \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}$$

Both $\mathbf{B}$ and $\mathbf{E}$ satisfy exactly the same differential equation of the following form:

$$\nabla^2 \psi = \mu_0 \epsilon_0 \frac{\partial^2 \psi}{\partial t^2}$$

Apparently $\mu_0 \epsilon_0$ has the dimension of $[T^2/L^2]$, and we therefore define $\sqrt{\mu_0 \epsilon_0} = c^{-1}$. Using the measured values $\mu_0 \approx 4\pi \times 10^{-7}$ N/A^2 and $\epsilon_0 \approx 8.854 \times 10^{-12}$ F/m gives $c = 2.998 \times 10^8$ m/s, the measured speed of light: light is an electromagnetic wave. This is how Maxwell identified optics as a branch of electromagnetism, more than two decades before Hertz produced these waves in 1887. To make things easier, let's first look at its 1D version:

$$\frac{\partial^2 \psi}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2}$$

Its general solution is:

$$\psi = \psi_1(x - ct) + \psi_2(x + ct)$$

and the first term corresponds to a wave moving in the x direction whereas the second term corresponds to a wave moving in the $-x$ direction, both at speed c. Thus, Maxwell's equations in vacuum admit waves of electromagnetic fields, namely **electromagnetic waves** or **EM waves**.

The electromagnetic spectrum. Electromagnetic waves span an enormous range of frequencies and wavelengths, from radio waves (roughly $\lambda > 0.1$ m) through microwaves, infrared, visible light (400–700 nm) and ultraviolet, to X-rays and gamma rays (roughly $\lambda < 10^{-11}$ m), as Fig. 14.2 shows. These band edges are one convention among several. Visible light occupies only a tiny portion of this spectrum. Remarkably, all these phenomena are governed by the same Maxwell equations.

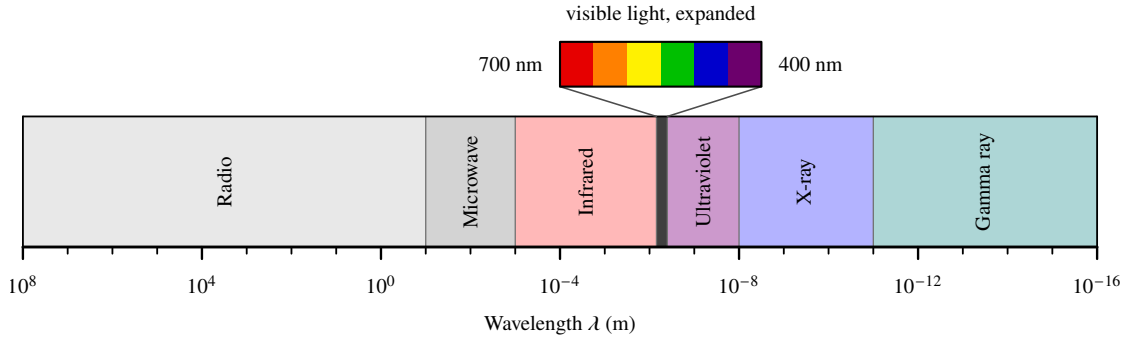


Figure 14.2: The electromagnetic spectrum. The wavelength axis spans 10^8 m to 10^{-16} m, from long radio waves down to gamma rays. Visible light occupies only a narrow window between 400 and 700 nm, far too narrow to label at this scale, so it is expanded in the inset above the bar.

We shall illustrate physical properties of the EM waves using plane-wave solutions (Fig. 14.3). Let's assume the plane-wave solution has the form

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_0 \cos[k(x - ct) + \phi]$$

$$\mathbf{B}(\mathbf{r}, t) = \mathbf{B}_0 \cos[k'(x - ct) + \phi']$$

For simplicity, we take the wave to travel in the x direction. For these plane-wave solutions, we have:

$$\nabla \cdot \mathbf{E} = -k\hat{x} \cdot \mathbf{E}_0 \sin[k(x - ct) + \phi]$$

$$\nabla \cdot \mathbf{B} = -k'\hat{x} \cdot \mathbf{B}_0 \sin[k'(x - ct) + \phi']$$

$$\nabla \times \mathbf{E} = -k\hat{x} \times \mathbf{E}_0 \sin[k(x - ct) + \phi]$$

Because the first two have to yield zero, we immediately know that $\hat{x} \cdot \mathbf{E} = \hat{x} \cdot \mathbf{B} = 0$, that is, both fields of an EM wave must be perpendicular to its direction of travel. In addition, we have:

$$\frac{\partial \mathbf{B}}{\partial t} = k'c\mathbf{B}_0 \sin[k'(x - ct) + \phi']$$

Because $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$ has to hold at any time and any position, we may take $k' = k$ and $\phi' = \phi$. Therefore, the magnetic and electric fields have the same frequency as well as the same phase, meaning they reach maximum or minimum at the same time and the same position. In addition, the amplitudes yield:

$$\hat{x} \times \mathbf{E}_0 = c\mathbf{B}_0$$

We note that $\omega = kc$ is the angular frequency of this wave, whereas $\mathbf{k} = k\hat{x}$ is its wave vector. Therefore, the above equation can also be written as:

$$\omega \mathbf{B}_0 = \mathbf{k} \times \mathbf{E}_0$$

or

$$\mathbf{B}_0 = \frac{1}{c}\left(\hat{k} \times \mathbf{E}_0\right)$$

Putting it back into the original expression, we also have:

$$\mathbf{B} = \frac{1}{c}\left(\hat{k} \times \mathbf{E}\right)$$

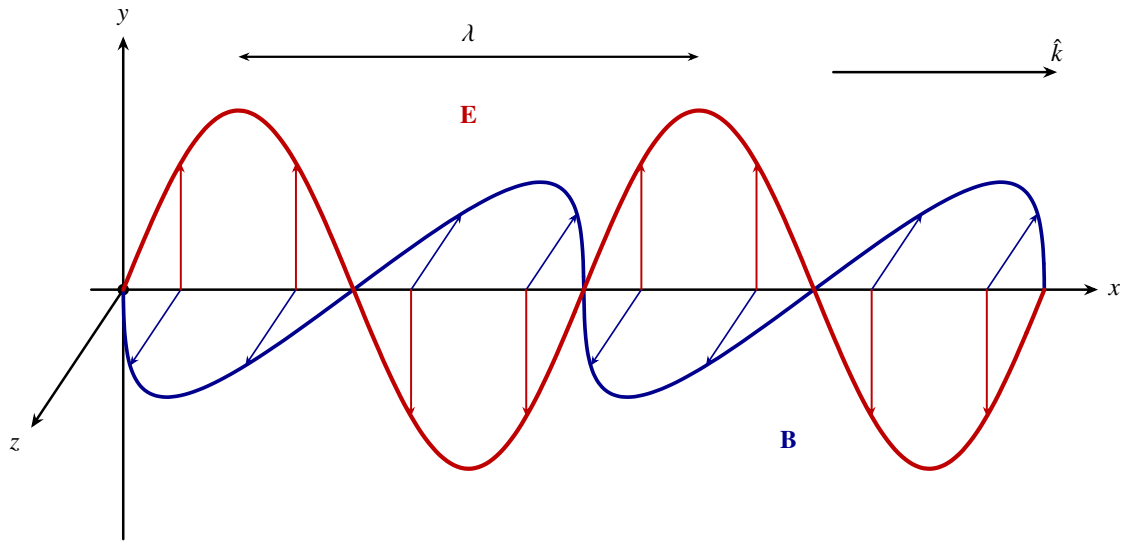


Figure 14.3: A traveling electromagnetic plane wave. The electric field **E**, magnetic field **B**, and propagation direction $\hat{k}$ are mutually perpendicular. **E** and **B** oscillate in phase with the same angular frequency ω. The wavelength λ and wave speed c are related by $c = \lambda f = \omega/k$.

14.3.1 Generation of Electromagnetic Waves

Electromagnetic waves are generated by accelerating charges. A practical scheme uses an LC oscillator to establish an angular frequency ω, which is coupled by a transformer and a transmission line to an antenna. The oscillating charges in the antenna produce changing electric and magnetic fields that propagate outward as electromagnetic waves at the same angular frequency ω. The simplest antenna is an electric dipole antenna, where charges oscillate back and forth along a conducting rod.

14.4 Energy Transport

The EM waves transport energy like other waves. We have seen that both the magnetic field and the electric field store energy. Their energy density expressions are:

$$u_E = \frac{1}{2}\epsilon_0 E^2$$

and

$$u_B = \frac{1}{2\mu_0} B^2$$

respectively.

Therefore, the total energy density of the field in vacuum is:

$$u = u_E + u_B = \frac{1}{2}\epsilon_0 E^2 + \frac{1}{2\mu_0}B^2$$

By analogy to charge conservation, you can see energy conservation also yields a continuity equation. In vacuum, where $\mathbf{j} = 0$, it says

$$\nabla \cdot \mathbf{S} + \frac{\partial u}{\partial t} = 0,$$

and wherever a current flows, the energy handed to the charges appears as a sink,

$$\nabla \cdot \mathbf{S} + \frac{\partial u}{\partial t} = -\mathbf{j} \cdot \mathbf{E},$$

where $\mathbf{S}$ represents *power per unit area*, or energy flow density. Therefore, if we calculate $\frac{\partial u}{\partial t}$ for an EM wave:

$$\begin{aligned}
\frac{\partial u}{\partial t} &= \frac{1}{2}\epsilon_0 \frac{\partial E^2}{\partial t} + \frac{1}{2\mu_0}\frac{\partial B^2}{\partial t} \\
&= \epsilon_0 \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t} + \frac{1}{\mu_0}\mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t} \\
&= \frac{1}{\mu_0}\mathbf{E} \cdot (\nabla \times \mathbf{B}) - \frac{1}{\mu_0}\mathbf{B} \cdot (\nabla \times \mathbf{E}) \\
&= -\frac{1}{\mu_0}\nabla \cdot (\mathbf{E} \times \mathbf{B})
\end{aligned}$$

The last step employs the identity $\nabla \cdot (\mathbf{C} \times \mathbf{D}) = \mathbf{D} \cdot (\nabla \times \mathbf{C}) - \mathbf{C} \cdot (\nabla \times \mathbf{D})$. Comparing this with the continuity equation fixes $\nabla \cdot \mathbf{S}$, so $\mathbf{S}$ is determined only up to the curl of some field; the Poynting vector is the natural choice:

$$\mathbf{S} = \frac{1}{\mu_0}\mathbf{E} \times \mathbf{B}$$

which is known as the **Poynting vector**. Applying this to the EM plane-wave, we have:

$$\mathbf{S} = \frac{1}{\mu_0}\mathbf{E}_0 \times \mathbf{B}_0 \cos^2[kx - \omega t + \phi]$$

For fast varying waves, we are only interested in its average, which is:

$$\bar{\mathbf{S}} = \frac{1}{2\mu_0}\mathbf{E}_0 \times \mathbf{B}_0$$

Its magnitude is $\bar{S} = E_0 B_0/(2\mu_0) = \frac{E_0^2}{2\mu_0 c} = \frac{1}{2}\epsilon_0 E_0^2 c$. $\frac{1}{2}\epsilon_0 E_0^2$ is the average energy density, while c is the speed of the EM wave, so $\bar{S}$ represents the energy transferred by the EM wave per unit area per unit time.

14.4.1 Intensity of Spherical Waves

When spherical wavefronts spread from an isotropic point source of power P_s, energy conservation requires that the intensity I (time-averaged power per unit area) decreases with distance r as:

$$I = \frac{P_s}{4\pi r^2}.$$

This inverse-square dependence is characteristic of any radiation spreading uniformly in three dimensions.

14.4.2 Radiation Pressure

Electromagnetic waves carry not only energy but also linear momentum. The momentum density of the field is $\mathbf{g} = \mathbf{S}/c^2$, so a single plane wave carries momentum across a surface perpendicular to it at the rate $\mathbf{S}/c$ per unit area. When an EM wave of intensity I falls perpendicularly on a surface:

- **Total absorption:** The radiation pressure is $p_r = I/c$, where I is the intensity.
- **Total reflection back along the path:** The momentum transfer is doubled, giving $p_r = 2I/c$.

Although radiation pressures are typically very small under ordinary conditions, focused laser beams can produce significant radiation pressure, enabling applications such as optical tweezers and laser cooling.

14.5 Exercises

1. A capacitor consists of two parallel circular plates, each of radius a. We connect the capacitor to an AC emf such that the charge on the capacitor follows

$$Q = Q_0 \sin \omega t.$$

Calculate the displacement current density j_d, the total displacement current i_d, and the magnetic field B at a distance $r < a$ from the axis, assuming $\omega a \ll c$. Here we ignore the boundary effect and assume the electric field inside the capacitor to be uniform.

2. An electromagnetic wave is traveling in the vacuum, where the electric field part is given by

$$\begin{aligned} E_x &= 0, \\ E_y &= 0.30 \cos\left[\omega\left(t - \frac{x}{c}\right)\right] \text{ V/m}, \\ E_z &= 0, \end{aligned}$$

where $\omega = 2\pi \times 10^8 \text{ s}^{-1}$.

Calculate the following properties of the EM wave:

   (a) The wavelength.

   (b) The propagation direction.

   (c) The magnitude and direction of the magnetic field.

3. A steady current i is flowing through a long cylindrical lead which has a radius a and resistivity ρ_e, and we assume that the current is uniformly distributed at the cross section and the electric field is also uniform inside the lead. Neglecting the magnetic field of the return circuit, calculate the following quantities:

   (a) The magnitudes and directions of $\mathbf{E}$, $\mathbf{B}$ and the Poynting vector $\mathbf{S}$ at a distance $r < a$ from the central axis.

   (b) For any point r inside the lead, verify the continuity equation

$$\nabla \cdot \mathbf{S} + \frac{\partial u}{\partial t} = -\rho_e J^2 = -\mathbf{j} \cdot \mathbf{E},$$

   where $J = |\mathbf{j}|$ is the magnitude of the current density.

4. A capacitor has a spherical shape, with inner shell radius r_1 and outer shell radius r_2. Initially it is at rest and a charge $+Q$ $(-Q)$ is uniformly distributed at the inner (outer) shell. A uniform magnetic field $\mathbf{B}$, produced by a long solenoid coaxial with the z axis that encloses the capacitor, points along $+z$ initially, and later it is slowly removed. The two shells are insulating, with the charges fixed on them; they are rigidly joined and can turn freely about the z axis, with moment of inertia I. Neglect the magnetic field produced by the rotating shells themselves. Find the final angular velocity, with its sense relative to $+z$.

5. A parallel-plate capacitor has circular plates of radius a a distance $d \ll a$ apart. The space between the plates is filled with a weakly conducting material of conductivity σ, whose permittivity may be taken as ϵ_0. At $t = 0$ the plates carry charges $+Q_0$ and $-Q_0$, and the capacitor is connected to nothing. Ignore the fringing field, so that the fields between the plates are uniform, and take the magnetic field between the plates to be azimuthal about the plates' axis.

(a) Find the charge $Q(t)$ on the positive plate for $t > 0$.

(b) Find the conduction current density and the displacement current density between the plates. Use the Ampère–Maxwell law to find the magnetic field between the plates.

(c) Find the total heat produced in the material, and compare it with the energy stored at $t = 0$. Using the Poynting vector, explain where that heat comes from.

6. Suppose that magnetic charge existed, with a density ρ_m and a current density $\mathbf{j}_m$, so that Gauss's law for magnetism became $\nabla \cdot \mathbf{B} = \mu_0 \rho_m$, and that magnetic charge were conserved, $\nabla \cdot \mathbf{j}_m + \partial \rho_m / \partial t = 0$. In this convention a point magnetic charge q_m, measured in A·m, at rest at the origin has the field $\mathbf{B} = \mu_0 q_m \hat{\mathbf{r}}/(4\pi r^2)$.

(a) Following the argument for $\mathbf{B}$ in Section 14.1, show that Faraday's law $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$ would then contradict the identity $\nabla \cdot (\nabla \times \mathbf{E}) = 0$, and that adding the term $-\mu_0 \mathbf{j}_m$ to its right-hand side removes the contradiction. Write the corresponding integral form for a fixed loop C bounding a fixed surface S, with the sense of C and the normal of S related by the right-hand rule.

(b) A thin ring of superconducting wire (zero resistance, so that $\mathbf{E} = 0$ inside the wire) with self-inductance L lies in the plane $z = 0$, centered on the z axis, and carries no current. A magnetic charge $q_m > 0$ moves along the z axis from far below the ring to far above it. Find the current in the ring after the charge has gone, and its sense: clockwise or counterclockwise, seen from above.

7. Two plane waves of the same amplitude E_0 and angular frequency ω travel through vacuum along the x axis in opposite directions. Their electric fields are $\mathbf{E}_1 = E_0 \cos(kx - \omega t)\, \hat{y}$ and $\mathbf{E}_2 = E_0 \cos(kx + \omega t)\, \hat{y}$, with $k > 0$ and $\omega = ck$, $c = 1/\sqrt{\mu_0 \epsilon_0}$.

(a) Using $\mathbf{B} = (\hat{k} \times \mathbf{E})/c$ for each wave, find the total fields $\mathbf{E}$ and $\mathbf{B}$. Find the positions x at which $\mathbf{E}$ is zero at all times and those at which $\mathbf{B}$ is zero at all times. At a position where neither is always zero, find the time from a maximum of $|\mathbf{E}|$ to the next maximum of $|\mathbf{B}|$.

(b) Find the Poynting vector of the total field, and its average over one period at each x. Verify that $\nabla \cdot \mathbf{S} + \partial u/\partial t = 0$.

(c) Suppose instead that the two waves traveled at an unknown speed $v = \omega/k$, each with $\mathbf{B} = (\hat{k} \times \mathbf{E})/v$. Show that the total field then satisfies Faraday's law for every v, but the Ampère–Maxwell law in vacuum only if $v = 1/\sqrt{\mu_0 \epsilon_0}$.

8. A fixed, infinitely thin sheet occupies the plane $x = 0$ in vacuum and carries no charge. An external source drives along it a uniform surface current $\mathbf{K}(t) = K_0 \cos \omega t\, \hat{y}$, where $K_0 > 0$ is the current per unit length (in A/m) crossing a line drawn on the sheet along $\hat{z}$. Consider the steady state, in which the sheet radiates plane waves away from itself on both sides, with $\mathbf{E}$ along $\hat{y}$ and $\mathbf{B}$ along $\hat{z}$, and there are no other fields.

(a) Apply the integral forms of Faraday's law and of the Ampère–Maxwell law to small fixed rectangles that cross the sheet, and show that E_y is continuous at the sheet while $B_z(0^+, t) - B_z(0^-, t) = -\mu_0 K_0 \cos \omega t$.

(b) Find $\mathbf{E}$ and $\mathbf{B}$ on both sides of the sheet.

(c) Find the time-averaged power per unit area carried away on each side. Show that together they equal the time-averaged power per unit area that the source spends driving the current against the electric field at the sheet, $-\overline{\mathbf{K} \cdot \mathbf{E}(0, t)}$.

9. A flat sail of area A reflects all the light that falls on it specularly, like a mirror. It is at rest in a parallel beam of light of intensity I (the time-averaged power per unit area through a surface perpendicular to the beam), with its normal at an angle θ to the beam, $0 \le \theta < \pi/2$.

   (a) Treat the incident and the reflected light as two beams, each carrying momentum along its direction at the rate of $\mathbf{S}/c$ per unit area of its cross-section (Section 14.4). Find the time-averaged force on the sail, magnitude and direction. Check the case $\theta = 0$ against the chapter.

   (b) The Sun radiates a power $L_\odot = 3.83 \times 10^{26}$ W isotropically and has mass $M_\odot = 1.99 \times 10^{30}$ kg; $G = 6.67 \times 10^{-11}$ N·m^2/kg^2. A sail facing the Sun ($\theta = 0$) at a distance r from it carries a total mass m, sail and payload. Find the value of m/A for which the light's force balances the Sun's gravity, show that it does not depend on r, and evaluate it.

   (c) Find the component of the force perpendicular to the beam, and the angle θ at which it is largest.